K-NN Text Classification of Web Pages

Results report - Statistical Modeling & Pattern Recognition, Project 3 (exercise 3.1) - November 10, 2026

Method

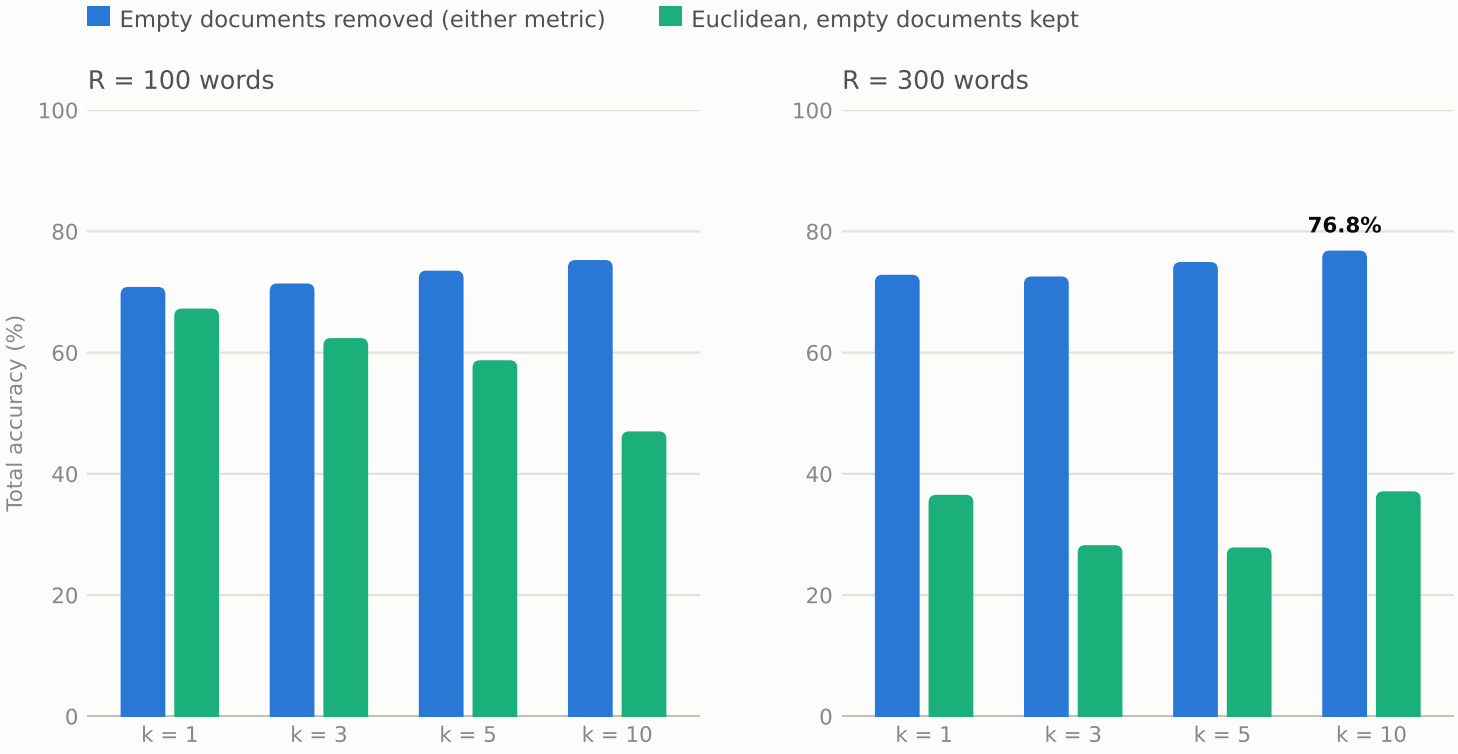
Four classes of university web pages (course, faculty, project, student; 1,396 documents, 4,749-word vocabulary) are represented by term-document matrices. Words are ranked by their log-entropy weight and the R most broadly distributed (lowest-weight) words are kept; the reduced matrix is TF-IDF weighted ((1 + log tf) x log(N/df)) and each document vector is normalized to unit length. Documents that contain none of the R words (all-zero vectors) are removed. A k-NN classifier is evaluated with 5-fold stratified cross-validation (random seed 0) for k = 1, 3, 5, 10, with Euclidean or cosine distance, for R = 100 and R = 300. For comparison, the same runs are repeated with the empty documents kept.

Results

		Empty documents removed		Empty documents kept	
R words	k neighbors	Euclidean	Cosine	Euclidean	Cosine
100	1	70.83%	70.83%	67.26%	69.55%
	3	71.41%	71.41%	62.39%	71.28%
	5	73.52%	73.52%	58.74%	72.99%
	10	75.27%	75.27%	47.00%	74.64%
300	1	72.84%	72.84%	36.54%	72.05%
	3	72.56%	72.56%	28.23%	71.20%
	5	74.95%	74.95%	27.86%	73.86%
	10	**76.83%**	**76.83%**	37.11%	76.94%

Total accuracy = mean accuracy over the 5 cross-validation folds. Best configuration in bold. Empty documents: 25 at R = 100, 15 at R = 300 (of 1,396).

Accuracy by number of neighbors



Observations

1. On clean data the two metrics are the same classifier. Every document vector has unit length, so squared Euclidean distance equals 2 - 2 x cosine similarity: both metrics rank the neighbors in the same order. With the empty documents removed they give exactly the same accuracy in every configuration.

2. Empty documents break Euclidean distance. 25 documents at R = 100 and 15 documents at R = 300 contain none of the selected words, so their vectors are all zeros. A zero vector lies at Euclidean distance exactly 1.0 from every unit vector, which is closer than any genuine neighbor whose cosine similarity is below 0.5 (distance sqrt(2 - 2 x similarity) > 1). At R = 300, 67% of the documents have no neighbor above 0.5 similarity, so their 'nearest neighbors' are the empty documents, whose labels span all four classes. Keeping them drops Euclidean accuracy at R = 300 to 28-37%. Because all the empty documents tie at the same distance, that figure also depends on how the sort breaks ties and is not reproducible across platforms. Cosine distance ranks a zero vector as maximally far and moves by at most 1.4 points.

3. More neighbors and more words both help, modestly. At R = 300 accuracy goes from 72.8% at k = 1 to 76.8% at k = 10 (R = 100: 70.8% to 75.3%). The increase is not strictly monotonic in k, and re-drawing the random folds moves a result by a standard deviation of up to 0.7 points (5 seeds), so differences of about one point between neighboring settings are not significant.

4. Word selection direction matters more than anything else. Selecting the 100 words with the highest entropy weight (rare, concentrated words) leaves 1,151 of 1,396 documents with empty vectors and yields 22.9% accuracy (cosine, k = 5, empty documents kept), below even the majority-class baseline. Selecting the lowest-weight (frequent, broadly distributed) words is what makes the results above possible.

Conclusion

For unit-normalized TF-IDF vectors the choice between Euclidean and cosine distance does not matter once documents with no selected words are removed; if they are kept, cosine is the robust choice and Euclidean fails. The best configuration found is R = 300 words with k = 10 neighbors, reaching 76.8% cross-validated accuracy on the four-class problem. The classes are unbalanced, so the fair baseline is always predicting the largest class ('student'), which scores 39.0%.